2

A ball is doing a projectile motion (egik atis) on a flat surface. The initial velocity of the ball is 40 m/s and the angle is 30 degree with the horizontal axis. What is the maximum height of the motion and what is the acceleration of the ball at the maximum height?
(4 Puan)

- $[61.2m\ ;\ -9.8\ \frac{m}{s^2}]$
- $[20.4m\ ;\ 0\ \frac{m}{s^2}]$
- $[61.2m\ ;\ 0\ \frac{m}{s^2}]$
- $[81.6m\ ;\ 0\ \frac{m}{s^2}]$
- $[20.4m\ ;\ -9.8\ \frac{m}{s^2}]$

3

A ball is doing a projectile motion (egik atis) on a flat surface. The initial velocity of the ball is 40 m/s and the angle is 30 degree with the horizontal axis. What is time of flight (havada kalma suresi) and what is the range (menzil) of the motion ?
(4 Points)

- $[7.07.\ s\ ;\ 141.39\ m]$
- none of these
- $[2.04.\ s\ ;\ 70.67\ m]$
- $[4.08\ s\ ;\ 141.39\ m]$
- $[8.08.\ s\ ;\ 323.2\ m]$

4

A car has a constant linear speed of 90 km/hr. The radius of its tire is 25cm. What is the angular speed of the tire?
(3 Points)

- 31.85 rad/s
- 100 rad/s
- 314 rad/s
- none of these
- 360 rad/s

5

A wheel is rotaing from its center of mass, its initial angular velocity is 0.25 rotation per second. Its angular acceleration is 0.5 rotation/ sec2 at t=0s. What is its angular velocity at 2 s ?
(3 Points)

- none of these
- 7.85 rad/s
- 1 rad/s
- 1.25 rad/s
- 3.14 rad /s

6

A wheel is rotaing from its center of mass, its initial angular velocity is 0.25 rotation per second. Its angular acceleration is 0.5 rotation/ sec2 at t=0s. What is the total angular displacement between 0s and 2s?
(4 Points)

- 9.42 rad
- 360 degrees
- 4.71 rad
- none of these
- 180 degrees

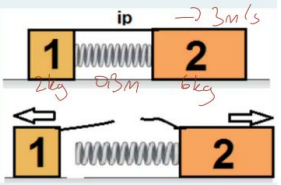
7

A wheel is rotaing from its center of mass, its initial angular velocity is 0.25 rotation per second. Its angular acceleration is 0.5 rotation/ sec2 at t=0s. What is the angular acceleration and radial acceleration at the edge of the wheel at t=2s? (Tekerin en dış noktasında açısal ivme ve radyal ivme nedir?)
(6 Points)

- [26.02 ; 1.18] m/s2
- [11.6 ; 0.5] m/s2
- [46.2 ; 2.36] m/s2
- [46.2 ; 3.14] m/s2
- [7.30 ; 1.57] m/s2

8

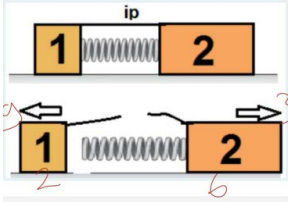
There is a spring attached to m2. Spring is compressed 0.3m. m1 and m2 are tied together with a rope (ip). m1=2kg, m2=6kg. There is no friction on the surface. When the rope is broken, the speed of m2 is 3m/s to the right as in the figure. What is the speed of m1 ?
(4 Points)



- none of these
- 0 m/s
- 1.67 m/s
- 3 m/s

9

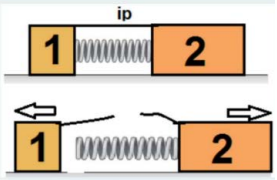
There is a spring attached to m2. Spring is compressed 0.3m. m1 and m2 are tied together with a rope (ip). m1=2kg, m2=6kg. There is no friction on the surface. When the rope is broken, the speed of m2 is 3m/s to the right as in the figure. What is the potential energy on the spring before the rope breaks?
(4 Points)



- None of these
- 27 J
- 108 J

10

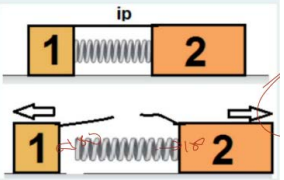
There is a spring attached to m2. Spring is compressed 0.3m. m1 and m2 are tied together with a rope (ip). m1=2kg, m2=6kg. There is no friction on the surface. When the rope is broken, the speed of m2 is 3m/s to the right as in the figure. What is the spring constant?
(4 Points)



- 800 N/m
- 1400 N/m
- 600 N/m
- 2400 N/m

11

There is a spring attached to m2. Spring is compressed 0.3m. m1 and m2 are tied together with a rope (ip). m1=2kg, m2=6kg. There is no friction on the surface. When the rope is broken, the speed of m2 is 3m/s to the right as in the figure. What is the spring force applied to m2 at the moment of the rope breaking (when the compression is still 0.3m) ?
(4 Points)



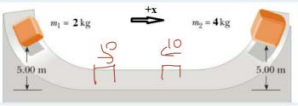
- 120 N
- 1440 N

12

There is a spring attached to m2. Spring is compressed 0.3m. m1 and m2 are tied together with a rope (ip). m1=2kg, m2=6kg. There is no friction on the surface. When the rope is broken, the speed of m2 is 3m/s to the right as in the figure. Which one(s) of the following statements are true?
(1) Just before the rope breaks the net force on the system is ZERO
(2) The total momentum before the rope breaks is equal to the total momentum after the rope breaks.
(3) When the rope breaks the accelerations of both masses are the same.
(4) The total energy is NOT conserved in this setup.
(5) At the instant of rope breaking, the spring force acting on m1 and m2 are NOT equal to each other.
(6) There is NOT enough information to solve this problem.
(4 Points)

13

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right after collision. (g=10 m/s2)
What is the velocity of the m1 and m2 respectively (sirasiyla), before the collision?
(4 Points)



- None of these
- [10 ; -10]m/s
- [20 ; -5]m/s
- [5 ; -20]m/s

14

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right AFTER collision. (g=10 m/s2)
If the velocity of m1 is V1 and the velocity of m2 is V2 AFTER the collision, which of the following relations are correct from momentum conservation?
(4 Points)

- $-10 = 4V_2 - 2V_1$
- $-10 = 2V_2 - V_1$
- $-20 = 2V_2 - V_1$
- $10 = 2V_2 - V_1$

15

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right AFTER collision. (g=10 m/s2)
If the velocity of m1 is V1 and the velocity of m2 is V2 AFTER the collision, which of the following relations are correct from energy conservation?
(4 Points)

- $200 = 2V_2^2 + V_1^2$
- $300 = 2V_2^2 - V_1^2$
- $600 = 2V_2^2 + V_1^2$
- $300 = 2V_2^2 + V_1^2$

16

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right AFTER collision. (g=10 m/s2)
If the velocity of m1 is V1 and the velocity of m2 is V2 AFTER the collision, what is the final SPEED of m1 (V1) AFTER the collision ?
(4 Points)

- $6.67\ \frac{m}{s}$
- $16.7\ \frac{m}{s}$
- $10\ \frac{m}{s}$
- $3.33\ \frac{m}{s}$

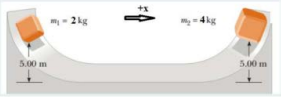
17

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right AFTER collision. (g=10 m/s2)
If the velocity of m1 is V1 and the velocity of m2 is V2 AFTER the collision, what is the final SPEED of m2 (V2) AFTER the collision ?
(4 Points)

- $6.67\ \frac{m}{s}$
- $13.3\ \frac{m}{s}$
- $16.7\ \frac{m}{s}$
- $10\ \frac{m}{s}$
- $3.33\ \frac{m}{s}$

18

There is NO friction on the surface. m1 =2kg and m2 =4kg are released from 5m and they collide ELASTICALLY. m1 moves left and m2 moves right AFTER collosion. (g=10 m/s2) and m1 goes up h1 height, m2 goes up h2 height. How high will each mass go AFTER the collision (h1 and h2)?
(6 Points)



- [13.89 ; 2.22] m
- [10 ; 2.5] m
- [2.22 ; 8.89] m
- [0.56 ; 13.89] m
- [13.89 ; 0.56] m

19

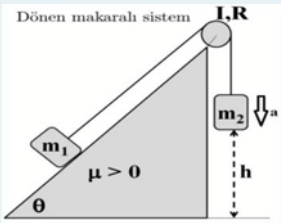
m1 and m2 are tied together with a string that goes over a pulley which has a moment of inertia of I and radius R. The string rotates the pulley without slipping. m2 is at a h distance and m2 goes down with an acceleration value of a. The inclined plane has theta angle and it has a friction coefficient of mu.

a) Draw free body diagram for all components of the system.

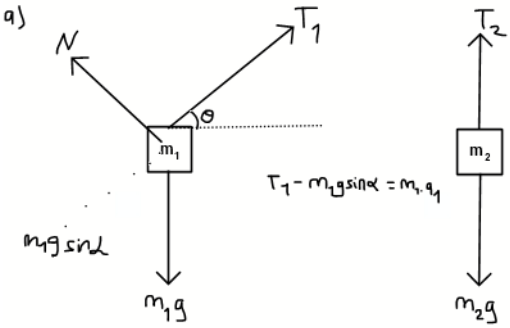
b) Are the tension on string are equal for both sides of pulley? If yes, explain why. If not equal, which one is bigger? and explain why.

c) find a acceleration expression in terms of g, m1, m2, I, R friction coefficient mu and theta angle.

d) Use energy conservation relation to find the final speed of m2 just before it hits the ground.
(Non-anonymous question)
(30 Points)



a)



$T_1 - m_1 g \sin\alpha = m_1 \cdot a_1$

$m_2 g - T_2 = m_2 \cdot a$

b)

c)

d) $U = m_2 \cdot g \cdot h$